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Improved Baselines with Visual Instruction Tuning

Haotian Liu[1] Chunyuan Li[2] Yuheng Li[1] Yong Jae Lee[1]
[1]University of Wisconsin–Madison [2]Microsoft Research, Redmond
https://llava-vl.github.io

Abstract

Large multimodal models (LMM) have recently shown encouraging progress with visual instruction tuning. In this paper, we present the first systematic study to investigate the design choices of LMMs in a controlled setting under the LLaVA framework. We show that the fully-connected vision-language connector in LLaVA is surprisingly powerful and data-efficient. With simple modifications to LLaVA, namely, using CLIP-ViT-L-336px with an MLP projection and adding academic-task-oriented VQA data with response formatting prompts, we establish stronger baselines that achieve state-of-the-art across 11 benchmarks. Our final 13B checkpoint uses merely 1.2M publicly available data, and finishes full training in ∼1 day on a single 8-A100 node. Furthermore, we present some early exploration of open problems in LMMs, including scaling to higher resolution inputs, compositional capabilities, and model hallucination, etc. We hope this makes state-of-the-art LMM research more accessible. Code and model will be publicly available.

1. Introduction

Large multimodal models (LMMs) have become increasingly popular in the research community, as they are the key building blocks towards general-purpose assistants [2, 30, 43]. Recent studies on LMMs are converging on a central concept known as visual instruction tuning [36]. The results are promising, *e.g.* LLaVA [36] and MiniGPT-4 [62] demonstrate impressive results on natural instruction-following and visual reasoning capabilities. To better understand the capability of LMMs, multiple benchmarks [17, 27, 34, 37, 55] have been proposed. Recent works further demonstrate improved performance by scaling up the pretraining data [3, 14, 54], instruction-following data [14, 18, 29, 58], visual encoders [3], or language models [39], respectively. The LLaVA architecture is also leveraged in different downstream tasks and domains, including region-level [8, 56] and pixel-level [26, 50] understanding, biomedical assistants [31], image generation [5], adversarial studies [6, 59].

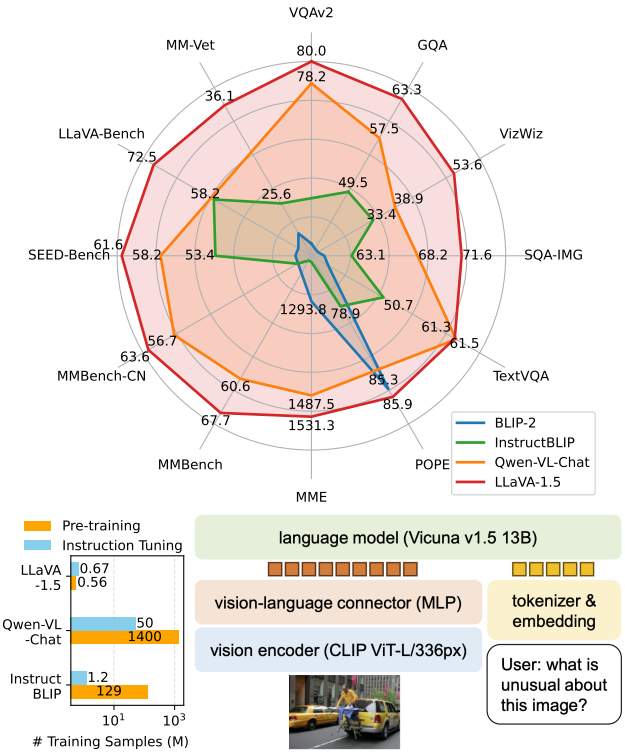


Figure 1. **LLaVA-1.5** achieves SoTA on a broad range of 11 tasks (Top), with high training sample efficiency (Left) and simple modifications to LLaVA (Right): an MLP connector and including academic-task-oriented data with response formatting prompts.

However, despite many benchmarks and developments, it still remains unclear what the best recipe is to train LMMs towards the goal of general-purpose assistants. For example, LLaVA [36] excels in conversational-style visual reasoning and even outperforms later approaches like InstructBLIP [14] on such benchmarks [55], while InstructBLIP excels in traditional VQA benchmarks that demands single-word or short answers. Given the significant differences in the model architecture and training data between them, the root cause of the disparity in their capabilities remains elusive, despite conjectures [37, 55]: the amount of training data, the usage of resamplers like Qformer [32], *etc*. To this

end, we present the first systematic study to investigate the design choices of LMMs in a controlled setting. Our study originates from LLaVA and builds a road map by carefully making effective contributions from the perspectives of the input, model, and data.

First, we unveil that the fully-connected vision-language connector in LLaVA is surprisingly powerful and data-efficient, and we establish stronger and more feasible baselines built upon the LLaVA framework. We report that two simple improvements, namely, an MLP cross-modal connector and incorporating academic task related data such as VQA, are orthogonal to the framework of LLaVA, and when used with LLaVA, lead to better multimodal understanding capabilities. In contrast to InstructBLIP [14] or Qwen-VL [3], which trains specially designed visual resamplers on hundreds of millions or even billions of image-text paired data, LLaVA uses one of the simplest architecture design for LMMs and requires only training a simple fully-connected projection layer on merely 600K image-text pairs. Our final model can finish training in $\sim$1 day on a single 8-A100 machine and achieves state-of-the-art results on a wide range of benchmarks. Moreover, unlike Qwen-VL [3] that includes in-house data in training, LLaVA utilizes only publicly available data.

Next, we delve into an early exploration of other open problems of large multimodal models. Our findings include: (1) Scaling to high-resolution image inputs. We show that LLaVA's architecture is versatile in scaling to higher resolutions by simply dividing images into grids and maintains its data efficiency; with the increased resolution, it improves the model's detailed perception capabilities and reduces hallucination. (2) Compositional capabilities. We find that large multimodal models are capable of generalizing to compositional capabilities. For example, training on long-form language reasoning together with shorter visual reasoning can improve the model's writing capability for multimodal questions. (3) Data efficiency. We show that randomly downsampling LLaVA's training data mixture by up to 75% does not significantly decrease the model's performance, suggesting that the possibility of a more sophisticated dataset compression strategy can further improve LLaVA's already efficient training pipeline. (4) Data scaling. We provide empirical evidence for the scaling of data granularity in conjunction with the model's capability is crucial for an improved capability without introducing artifacts like hallucination.

In sum, we perform a systematic study on the training of large multimodal models, and introduce a simple yet effective approach to balance the multitask learning and effective scaling for large multimodal models. Our improved baselines, LLaVA-1.5, uses only *public* data, achieves the state-of-the-art on a broad range of 11 tasks, and is significantly more data-efficient than previous approaches. By rethinking the conventional approaches and exploring the open problems in visual instruction tuning, we pave the way for more robust and capable systems for LMMs. We hope these improved and easily-reproducible baselines will provide a reference for future research in open-source LMMs.

2. Related Work

Instruction-following large multimodal models (LMMs). Common architectures include a pre-trained visual backbone to encode visual features, a pre-trained large language model (LLM) to comprehend the user instructions and produce responses, and a vision-language cross-modal connector to align the vision encoder outputs to the language models. As shown in Fig. 1, LLaVA [36] is perhaps the simplest architecture for LMMs. Optionally, visual resamplers (*e.g.* Qformer [32]) are used to reduce the number of visual patches [3, 14, 62]. Training an instruction-following LMM usually follows a two-stage protocol. First, the vision-language alignment pretraining stage leverages image-text pairs to align the visual features with the language model's word embedding space. Earlier works utilize relatively few image-text pairs (*e.g.* $\sim$600K [36] or $\sim$6M [62]), while some recent works pretrain the vision-language connector for a specific language model on a large amount of image-text pairs (*e.g.* 129M [14] and 1.4B [3]), to maximize the LMM's performance. Second, the visual instruction tuning stage tunes the model on visual instructions [36], to enable the model to follow users' diverse requests on instructions that involve the visual contents. Dealing with higher resolution with grids in LMM are studied in con-current works [1, 28, 53].

Multimodal instruction-following data. In NLP, studies show that the quality of instruction-following data largely affects the capability of the resulting instruction-following models [61]. For visual instruction tuning, LLaVA [36] is the pioneer to leverage text-only GPT-4 to expand the existing COCO [35] bounding box and caption dataset to a multimodal instruction-following dataset that contains three types of instruction-following data: conversational-style QA, detailed description, and complex reasoning. LLaVA's pipeline has been employed to expand to textual understanding [57], million-scales [58], and region-level conversations [8]. InstructBLIP [14] incorporates academic-task-oriented VQA datasets to further enhance the model's visual capabilities. Conversely, [7] identifies that such naive data merging can result in models that tend to overfit to VQA datasets and thus are unable to participate in natural conversations. The authors further propose to leverage the LLaVA pipeline to convert VQA datasets to a conversational style. While this proves effective for training, it introduces added complexities in data scaling. However, in NLP, the FLAN family [13, 51] shows that adding a large number of academic language tasks for instruction tuning can effectively improve the gen-